RoboPIN

Grounded Embodied Reasoning via Pinned Chain-of-Thought

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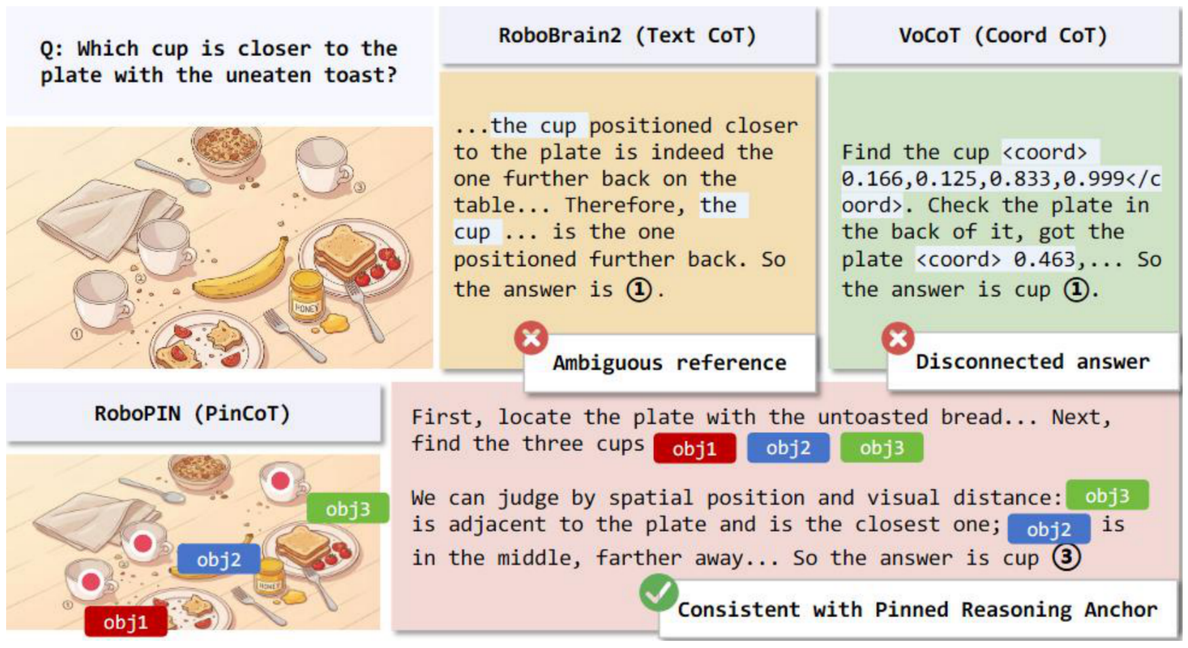
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Motivation



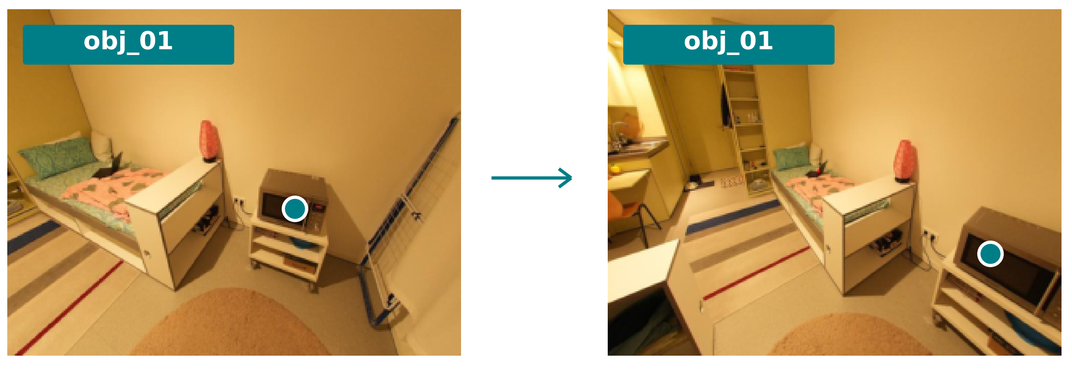
1 Reference drift

Entity references can switch between similar objects across reasoning steps.

2 Reasoning–answer misalignment

The final answer can point to a different entity from the one analyzed.

Key Insight: Persistent Identity



One identity, view-specific grounding

`obj_01 · view 0 · point [634, 576]`

`obj_01 · view 1 · point [844, 706]`

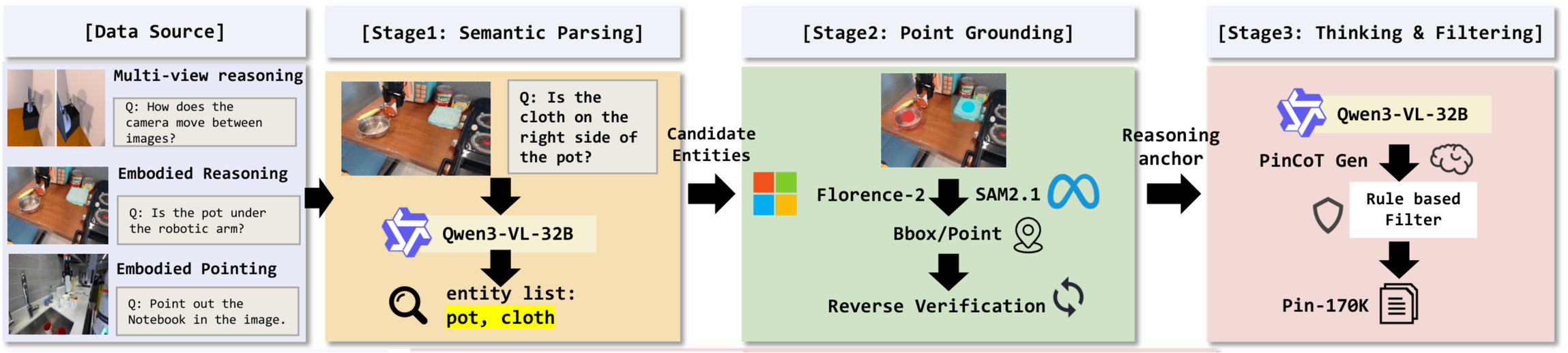
Track identity. Verify grounding.

Introduce each entity with a full anchor; reuse its ID and update view-specific coordinates.

QR PLACEHOLDER

PAPER / CODE

Our Method: RoboPIN / PinCoT

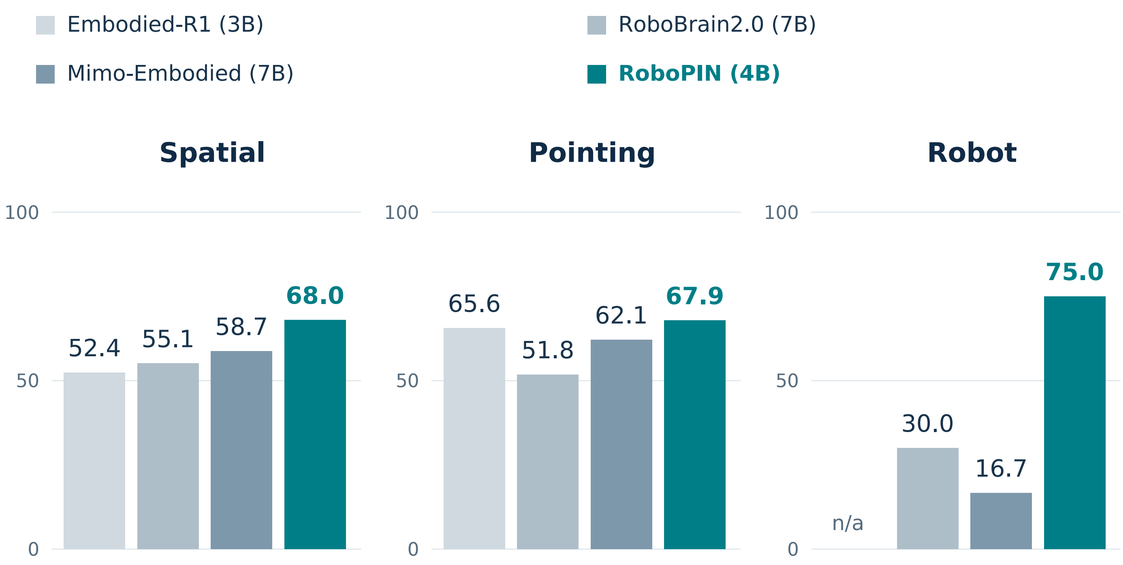


Three-stage post-training

SFT — Embodied knowledge → **CoT-SFT** — Structured reasoning → **RFT / GRPO** — Process alignment

Process-supervised RFT rewards format, identity consistency, pin localization and final-answer correctness.

Experimental Results



Scores (%): spatial mean over 13 columns; pointing mean over 5 benchmarks; robot success over 60 trials.

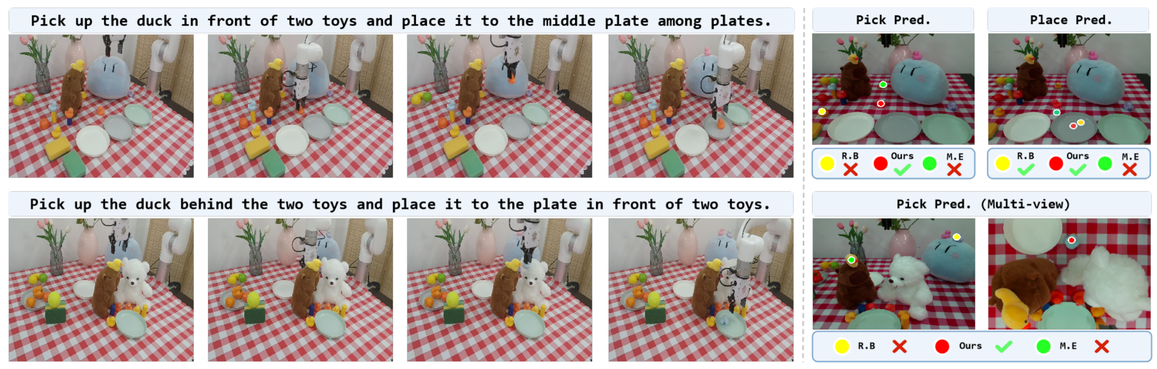
Selected benchmarks · score (%)

Benchmark	Mimo-7B	RoboPIN	Δ pp
CrossPoint	36.8	**72.2**	**+35.4**
BLINK multi-view	26.3	**72.9**	**+46.6**
EmbSpatial	76.2	**83.1**	**+6.9**

18.8% → 99.9% cross-step ID coverage

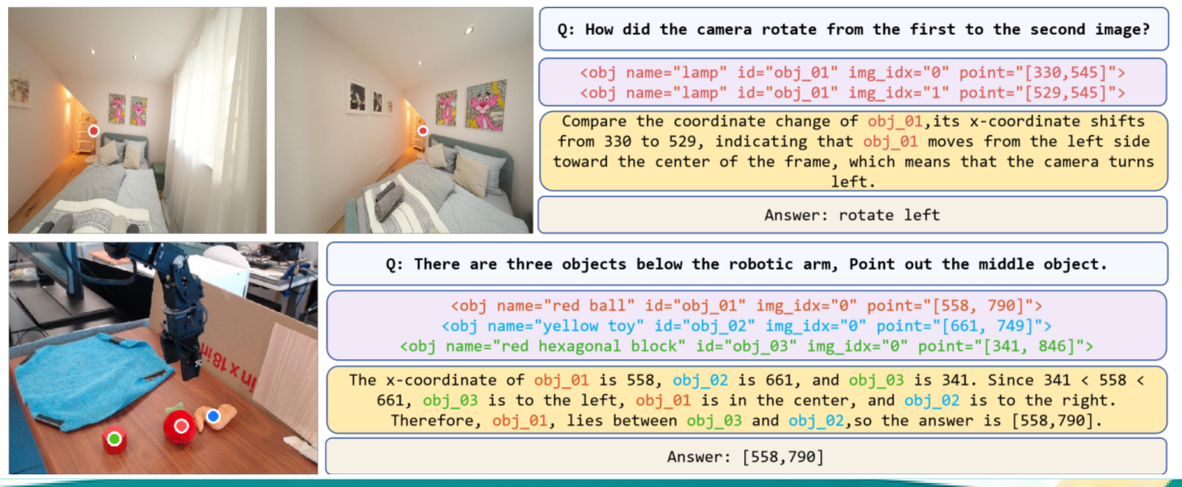
Without RFT → RoboPIN · EmbSpatial

Qualitative Results



Zero-shot xArm · successful executions / 20

20 / 20	14 / 20	11 / 20
Spatial	Multi-view	Long-horizon



Persistent visual anchors make embodied reasoning traceable.